Week 03 Reading: Inference and resampling (nothing to submit)

Likelihood, Bayes, the probability integral transform, and the bootstrap

2026-09-09

No Monday class (Labor Day), so reading questions are due Tuesday 9/8 at 11:59 pm on Canvas.

Assigned

1. Chapters 1 through 3 of Downey (2021), which build from probability through Bayes's theorem to working with distributions as priors and posteriors. The book is free online. If you already know Bayes's theorem well, skim chapters 1 and 2 and read chapter 3 carefully.
2. Sections 3.1 and 3.2 of Mudelsee (2010), which introduce the bootstrap in a climate context and explain why a naïve bootstrap fails on a record with autocorrelation. This reading is available through the Rice library; search the ISBN 978-90-481-9936-4 at library.rice.edu.

Further reading

1. Chapter 2 of Davison & Hinkley (1997) for the formal treatment of bootstrap theory. This is the standard reference; read it if you want the proofs behind what we do on Friday.
2. John Rauser, "Statistics Without the Agonizing Pain," Strata 2014. A 25-minute talk that builds the bootstrap from scratch with no equations. Search the title on YouTube.

Questions

Write a few sentences on each.

1. In the Downey reading, the prior is updated by data to produce a posterior. In your own words, what happens to the posterior as you collect more data: does the prior matter more or less?
2. A colleague fits an AR(1) model to a 30-year temperature record and bootstraps the lag-1 coefficient by resampling individual years with replacement. What assumption does that bootstrap make, and why might it give confidence intervals that are too narrow?
3. The Mudelsee reading introduces the block bootstrap as a fix for dependent data. What does the block length control, and what goes wrong if the blocks are too short or too long?

Bibliography

Davison, A. C., & Hinkley, D. V. (1997). *Bootstrap Methods and Their Application* (Reprinted with corr, Vol. 5.Series Number 1). New York: Cambridge University Press. https://doi.org/10.1017/CBO9780511802843

Downey, A. B. (2021). *Think Bayes*. "O'Reilly Media, Inc.". Retrieved from https://allendowney.github.io/ThinkBayes2/

Mudelsee, M. (2010). *Climate Time Series Analysis: Classical Statistical and Bootstrap Methods* (1 ed. 2010.). Dordrecht: Springer Netherlands. https://doi.org/10.1007/978-90-481-9482-7